

To find the value of $x + 2016z$, we first need to determine the conditions under which a triangle is n -good and n is amazing.

1. Finding the ratio $\frac{CP}{CH}$: Let B be the origin $(0,0)$ and BC be the positive x -axis. Since $\angle B = 75^\circ$, AB makes an angle of 75° with BC . Using coordinate geometry and the properties of the orthocenter H , the angle between BH and BC is $90^\circ - \gamma$. H_A , the reflection of H across BC , has coordinates $(r \sin \gamma, -r \cos \gamma)$, where $r = BH$. The point X on the opposite ray of BC with $BX = r$ is $(-r, 0)$. The line XH_A intersects CH at P . By calculating the intersection, the ratio of the directed lengths is found to be:

$$\frac{CP}{CH} = \frac{\tan 15^\circ + \tan(45^\circ - \gamma/2)}{\tan 15^\circ - \tan(45^\circ - \gamma/2)} = \frac{\sin \theta}{\sin(30^\circ - \theta)}$$

where $\theta = 7.5^\circ + \alpha/2$. We are given that $\frac{CP}{CH} = \sqrt{n}$, yielding $\cot \theta = \frac{2}{\sqrt{n}} + \sqrt{3}$.

2. Evaluating the lengths AQ and H_CQ : Following a similar procedure with AB as the x -axis, we find the coordinates of H_C and Q . The lengths of the segments AQ and H_CQ are determined by the geometry of the triangle. The ratio of these lengths simplifies remarkably to:

$$q = \frac{AQ}{H_CQ} = 2 \sin(30^\circ - \theta)$$

Substituting $\sin \theta = \frac{1}{\sqrt{1+\cot^2 \theta}}$ from our previous deduction, we find:

$$q = \frac{1}{\sqrt{n+1+\sqrt{3n}}}$$

For H_CQ and AQ to both be integers (which defines n as *amazing*), their ratio q must be a rational number. Thus, $n+1+\sqrt{3n}$ must be the square of a rational number.

3. Solving for amazing n : For $\sqrt{3n}$ to be rational, n must be of the form $n = 3m^2$ for some positive odd integer m . This forces $3m^2 + 3m + 1 = R^2$ for some integer R . Multiplying by 4 gives:

$$3(2m+1)^2 + 1 = (2R)^2 \implies X^2 - 3Y^2 = 1$$

where $X = 2R$ and $Y = 2m+1$. Since m is odd, $Y \equiv 3 \pmod{4}$. The solutions for Y in the Pell equation $X^2 - 3Y^2 = 1$ are generated by $Y_k = \frac{(2+\sqrt{3})^k - (2-\sqrt{3})^k}{2\sqrt{3}}$. $Y \equiv 3 \pmod{4}$ occurs for indices $k = 4i - 1$ for $i \geq 1$. Hence, the i -th amazing integer is $n_i = 3 \left(\frac{Y_{4i-1}-1}{2} \right)^2$.

4. Finding the growth factors x and z : We are interested in the behavior of n_{2i} for sufficiently large i :

$$n_{2i} \approx \frac{3}{4} Y_{8i-1}^2 \approx \frac{3}{4} \left(\frac{(2+\sqrt{3})^{8i-1}}{2\sqrt{3}} \right)^2 = \frac{1}{16(2+\sqrt{3})^2} \left((2+\sqrt{3})^{16} \right)^i$$

Thus, the base of the exponential growth is $x = (2+\sqrt{3})^{16}$, and the leading constant supremum z is $z = \frac{1}{16(2+\sqrt{3})^2} = \frac{7-4\sqrt{3}}{16}$.

5. Final Calculation: We evaluate x and $2016z$:

$$x = (2+\sqrt{3})^{16} = 708158977 + 408855936\sqrt{3}$$

$$2016z = 2016 \left(\frac{7-4\sqrt{3}}{16} \right) = 126(7-4\sqrt{3}) = 882 - 504\sqrt{3}$$

Summing these gives:

$$x + 2016z = 708159859 + 408855432\sqrt{3}$$

Here we have $p = 708159859$, $q = 408855432$, and $r = 3$ (which is square-free).

The sum $p + q + r$ is:

$$708159859 + 408855432 + 3 = 1117015294$$

Final Answer: $p + q + r = 1117015294$